

Cyber Cafe System Design

Architecture, routes, data structures, and operational flow

1. High-Level Architecture

Component	Description
Browser Client	Admin and user interfaces rendered from Flask templates
Flask Server	Handles routing, session logic, OAuth, POS, and payment management
Upload Storage	Local filesystem storage for payment screenshots
Third-party OAuth	Google accounts OAuth 2.0 authorization
QR Generation	UPI payment payload encoded as QR image URL

2. Route Design

Route	Purpose
/	Login page and local credential authentication
/login/google	Start Google OAuth
/auth/google/callback	Receive Google OAuth callback
/admin	Admin station dashboard
/admin/pos	Retail POS view
/admin/pos/checkout	QR payment screen
/admin/pos/payment	Upload payment screenshot
/admin/payments	Payment verification queue
/user	User dashboard
/api/stations	JSON station status API

3. Data Structures

- USERS: predefined login accounts and roles
- stations: station state, active session, timing, and charge
- retail_items: available products and services
- customers: leaderboard and user summary data
- activity_log: recent system actions

- payment_requests: uploaded proof submissions and verification state

4. Operational Flows

4.1 Admin Station Flow

- Admin logs in
- Admin selects station and duration
- Session timer starts and dashboard updates remaining time
- Admin can release the station manually

4.2 Payment Flow

- Admin adds items to cart
- System calculates total and creates a payment reference
- QR page shows amount and payment receiver details
- User pays externally and screenshot is uploaded
- Admin verifies or rejects the proof submission

5. Security and Access Notes

- Session-based access control is used throughout the app
- Admin-only routes are protected by admin_required
- Google OAuth uses state tokens to mitigate CSRF during sign-in
- Payment screenshot upload is limited by allowed file extension
- Production deployment should use a regenerated secret key and HTTPS